Pricing of a novel parametric weather derivative - A Multi-variate Spatial Basket

HKUST

Workflow

1. Local Risk Exposure - Baseline Stochastic Models Fitted to local Data: $S = f(Y_l(t))$
   - Weather variables and Climate events - Class of Gauss-Markov processes suitable for indifference pricing measure.
   - Continuous-time AR(p) model, Seasonal volatility fit.
   - Generalized Hyperbolic Distribution fit.
   - Normal Inverse Gaussian fit.
2. Global Traded Index - Futures / Parametric Proxy: $Y_g(t)$
   - Constructing Real Estate investment portfolio and HDD, CDD, CAT futures index proxy under risk-neutral distribution.
   - Climate data for Compound Risk Indices.
3. Pricing + Hedging - Spatial Basket: $X(t)$ and $a(t)$ Strategy
   - Constructing synthetic future index.
   - Spatio-temporal climate model.

Biovariables and Morphology params

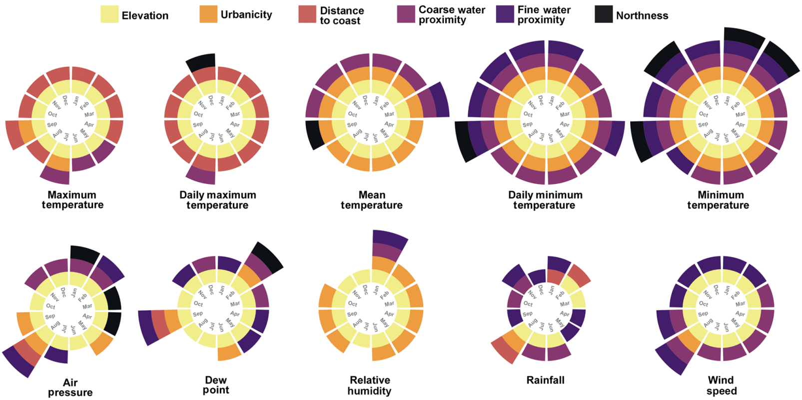

Figure: Reference variables

Real Estate Investment and weather index - Y_l; Y_g

- Exposure model (since no market portfolio) Real Estate - Estate-wise aggregation: local climate exposure Y_l

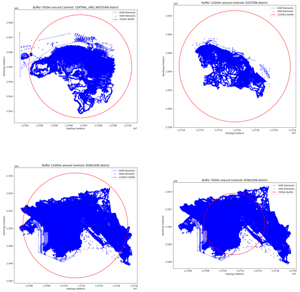

Figure: Intra-District aggregation

Futures on HDD, CDD, CAT, and PRIM - usual traded indices. Multivariate Index strategy based on a linearization based feedback law : get synthetic global climate index Y_g

Pricing and Model Specification - no basis risk

$$F(t,y)\left(\alpha g(t,y)+\frac{1}{2}\frac{\mu^2(y)}{\sigma^2(y)}\right)=\frac{\partial F}{\partial t}(t,y)+GF$$

with terminal condition $F(T,y)=1$. We decompose the state variable Y by using proxies of y: Y_1 as local risk and Y_2 as global risk (for payoff). Using Feynman-Kac -

$$F(t,Y_1,Y_2)=\mathbb{E}_y\left[\exp\left(-\int_t^T\left(\alpha g(s,Y_2)+\frac{1}{2}\frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))}\right)ds\right)\right].$$

E_y - over joint distribution Y_1 and Y_2 s.t. inf generator G accounts for coupling terms. with $V(t,x,y)=-e^{-\alpha x}F(t,Y_1,Y_2)$

General Solution (Carmona 2005) -

$$p=\frac{1}{\alpha}\ln\left(\frac{\mathbb{E}_y\left[\exp\left(-\int_0^T\frac{1}{2}\frac{\mu^2(y_s)}{\sigma^2(y_s)}\,ds\right)\right]}{\mathbb{E}_y\left[\exp\left(-\int_0^T\left(\alpha g(s,\hat{Y}_s)+\frac{1}{2}\frac{\mu^2(y_s)}{\sigma^2(y_s)}\right)\,ds\right)\right]}\right)$$

Pricing and Model Specification - basis risk

Unhedgable basis/residual risk or when coupling effect is negligible/can be treated as a perturbation to the independent case -
With terminal condition $F(T, Y_1, Y_2) = 1$

$$F(t, Y_1, Y_2)\left(\alpha g(t, Y_2) + \frac{1}{2}\frac{\mu^2(Y_1)}{\sigma^2(Y_1)}\right) = \frac{\partial F}{\partial t}(t, Y_1, Y_2) + \mathcal{L}_{Y_1}F + \mathcal{L}_{Y_2}F$$

$$F(t, Y_1, Y_2) = \mathbb{E}_{Y_1, Y_2}\left[\exp\left(-\int_t^T \left(\alpha g(s, Y_2(s)) + \frac{1}{2}\frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))}\right) ds\right)\right]$$

$$= \mathbb{E}_{Y_1}\left[\exp\left(-\int_t^T \frac{1}{2}\frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))} ds\right)\right] \times \mathbb{E}_{Y_2}\left[\exp\left(-\int_t^T \alpha g(s, Y_2(s)) ds\right)\right]$$

Trading strategy: $\phi_t = \frac{\mu(Y_1)}{(\sigma^2(Y_1)\alpha)}$ — Case ignored since p = 0

Pricing and Model Specification - basis risk

Simplified coupled case (b/w local and global risk) - Price through the 'distortion solution' -

$$p = \frac{\ln \mathbb{E}\left[e^{(1-\rho^2)\Psi(Y_2(T))\frac{d\mathbb{Q}}{d\mathbb{P}}} \right]}{\alpha(1-\rho^2)}, \quad \frac{d\mathbb{Q}}{d\mathbb{P}} = e^{-\frac{1}{2}\int_0^T (\mu_1^2/\sigma_1^2)\, d\tau + \int_0^T \frac{\mu_1}{\sigma_1} dW_t^S}$$

$$\phi_t = \frac{\rho}{\sigma(Y_1)} a(Y_2)p' + \frac{\mu(Y_1)}{\sigma(Y_1)^2\alpha}.$$

where -

$$dS_t = \mu(Y_1, t)dt + \sigma(Y_1, t)dW_t^s$$

$$dY_t = b(Y_2, t)dt + a(Y_2, t)(\rho dW_t^s + \sqrt{1-\rho^2} dW_t^{s\perp})$$

$$W^{s\perp} = \frac{1}{\sqrt{1-\rho^2}} W^y - \frac{\rho}{\sqrt{1-\rho^2}} W^s$$

Pricing and Model Specification - basis risk

Proposed coupling dynamics - adv-diff-reac system

$$\frac{\partial y_1}{\partial t} + w \cdot \nabla y_1 - \nabla \cdot (\epsilon \nabla y_1) = f_1 - K y_1 y_2$$

$$\frac{\partial y_2}{\partial t} + w \cdot \nabla y_2 - \nabla \cdot (\epsilon \nabla y_2) = f_2 - K y_1 y_2$$

$$\frac{\partial u}{\partial t} + w \cdot \nabla u - \nabla \cdot (\epsilon \nabla u) = f_3 - K y_1 y_2 - K u$$

$\mathrm{A} + \mathrm{B} \rightarrow C.$

Temporal Dynamics of State Variables

Temperature example - for basis risk as the source of market incompleteness

$$dT(t,x) = d\Lambda(t,x) - \alpha(x)(T(t,x) - \Lambda(t,x))\,dt + \sigma(t,x)\,dW(t,x)$$

- $W(t,\cdot)$: $L^2(D)$-valued Wiener process
- Continuous spatial covariance function $q(x,y)$, strictly positive definite and symmetric.
- Given weather index $I(x_i)$ at locations $x_1,\dots,x_n$.
- Optimal strategy $a(t)$ minimizing variance of error in replicating $I(y)$ at location y using spatial correlation and geometry of locations.
- Risk Neutral Time-Dynamics of Index Futures Prices;

$$dF_{\text{index}}(t,\tau_1,\tau_2) = \sigma(t)\int_{\tau_1}^{\tau_2} e_1' \exp(A(s-t))e_p\,\Phi$$

Φ is the cumulative standard normal distribution

Fitting the Model

- Explicit solution of $X(t)$:

$$X(s) = \exp(A(s-t))x + \int_t^s \exp(A(s-u))e_p\sigma(u)\, dB(u)$$

- Temperature $T(t)$ is defined as:

$$T(t) = \Lambda(t) + X_1(t)$$

- $X_1(t)$: CAR(p) model, $\Lambda(t)$: seasonality function

1. Fit seasonal function $\Lambda(t)$ with least squares.
2. Fit AR(p) model on deseasonalized temperatures.
3. Fit seasonal volatility $\sigma(t)$ to residuals.

Contract Payoff and Pricing

Utility with a derivative written on the underlying process Y - akin to Asian structure and $K = 0$
$f_1(Y) = \left(\int_0^T Y_t\, dt - K\right)^+; f_2(Y) = \left(\int_0^T \not\Vdash_{\{Y_t > \epsilon\}} dt - K\right)^+,$

$$f(Y) = \int_0^T h(Y_s)\, ds, \quad 0 \le t' \le t \le T.$$

- Numerical solution of underlying optimization problem analysed through its dual under HJB (David)
- Explicit closed form solution to HJB: RN measure vs historical measure vs Indifference measure (Musiela)

Spatial Hedging - Weight Matrix - Global index Y_g

a)

LCZ	1	2	3	4	5	6	7	8	9	10	A	B	C	D	E	F	G
1	0.00	0.08	0.17	0.33	0.42	0.50	0.42	0.25	0.58	0.25	0.67	0.75	0.92	1.00	0.75	1.00	1.00
2	0.08	0.00	0.08	0.42	0.33	0.42	0.33	0.17	0.58	0.17	0.58	0.67	0.75	0.92	0.67	0.92	0.92
3	0.17	0.08	0.00	0.50	0.42	0.33	0.25	0.08	0.42	0.08	0.50	0.58	0.67	0.83	0.58	0.83	0.83
4	0.33	0.42	0.50	0.00	0.08	0.17	0.25	0.50	0.25	0.42	0.50	0.42	0.50	0.67	0.92	0.67	0.67
5	0.42	0.33	0.42	0.08	0.00	0.08	0.17	0.33	0.17	0.33	0.42	0.33	0.50	0.58	0.83	0.58	0.58
6	0.50	0.42	0.33	0.17	0.08	0.00	0.08	0.25	0.08	0.25	0.33	0.25	0.33	0.42	0.75	0.50	0.50
7	0.42	0.33	0.25	0.25	0.17	0.08	0.00	0.33	0.17	0.33	0.25	0.33	0.42	0.58	0.83	0.58	0.58
8	0.25	0.17	0.08	0.50	0.33	0.25	0.33	0.00	0.25	0.08	0.58	0.50	0.58	0.75	0.50	0.67	0.67
9	0.58	0.58	0.42	0.25	0.17	0.08	0.17	0.25	0.00	0.33	0.42	0.33	0.25	0.42	0.67	0.42	0.42
10	0.25	0.17	0.08	0.42	0.33	0.25	0.33	0.08	0.33	0.00	0.58	0.50	0.58	0.75	0.50	0.67	0.67
A	0.67	0.58	0.50	0.50	0.42	0.33	0.25	0.58	0.42	0.58	0.00	0.08	0.17	0.33	0.58	0.33	0.33
B	0.75	0.67	0.58	0.42	0.33	0.25	0.33	0.50	0.33	0.50	0.08	0.00	0.08	0.25	0.50	0.25	0.25
C	0.92	0.75	0.67	0.50	0.50	0.33	0.42	0.58	0.25	0.58	0.17	0.08	0.00	0.17	0.42	0.17	0.17
D	1.00	0.92	0.83	0.67	0.58	0.42	0.58	0.75	0.42	0.75	0.33	0.25	0.17	0.00	0.25	0.08	0.08
E	0.75	0.67	0.58	0.92	0.83	0.75	0.83	0.50	0.67	0.50	0.58	0.50	0.42	0.25	0.00	0.25	0.25
F	1.00	0.92	0.83	0.67	0.58	0.50	0.58	0.67	0.42	0.67	0.33	0.25	0.17	0.08	0.25	0.00	0.08
G	1.00	0.92	0.83	0.67	0.58	0.50	0.58	0.67	0.42	0.67	0.33	0.25	0.17	0.08	0.25	0.08	0.00

b)

LCZ	1	2	3	4	5	6	7	8	9	10	A	B	C	D	E	F	G
1	1.00	0.92	0.83	0.67	0.58	0.50	0.58	0.75	0.42	0.75	0.33	0.25	0.08	0.00	0.25	0.00	0.00
2	0.92	1.00	0.92	0.58	0.67	0.58	0.67	0.83	0.42	0.83	0.42	0.33	0.25	0.08	0.33	0.08	0.08
3	0.83	0.92	1.00	0.50	0.58	0.67	0.75	0.92	0.58	0.92	0.50	0.42	0.33	0.17	0.42	0.17	0.17
4	0.67	0.58	0.50	1.00	0.92	0.83	0.75	0.50	0.75	0.58	0.50	0.58	0.50	0.33	0.08	0.33	0.33
5	0.58	0.67	0.58	0.92	1.00	0.92	0.83	0.67	0.83	0.67	0.58	0.67	0.50	0.42	0.17	0.42	0.42
6	0.50	0.58	0.67	0.83	0.92	1.00	0.92	0.75	0.92	0.75	0.67	0.75	0.67	0.58	0.25	0.50	0.50
7	0.58	0.67	0.75	0.75	0.83	0.92	1.00	0.67	0.83	0.67	0.75	0.67	0.58	0.42	0.17	0.42	0.42
8	0.75	0.83	0.92	0.50	0.67	0.75	0.67	1.00	0.75	0.92	0.42	0.50	0.42	0.25	0.50	0.33	0.33
9	0.42	0.42	0.58	0.75	0.83	0.92	0.83	0.75	1.00	0.67	0.58	0.67	0.75	0.58	0.33	0.58	0.58
10	0.75	0.83	0.92	0.58	0.67	0.75	0.67	0.92	0.67	1.00	0.42	0.50	0.42	0.25	0.50	0.33	0.33
A	0.33	0.42	0.50	0.50	0.58	0.67	0.75	0.42	0.58	0.42	1.00	0.92	0.83	0.67	0.42	0.67	0.67
B	0.25	0.33	0.42	0.58	0.67	0.75	0.67	0.50	0.67	0.50	0.92	1.00	0.92	0.75	0.50	0.75	0.75
C	0.08	0.25	0.33	0.50	0.50	0.67	0.58	0.42	0.75	0.42	0.83	0.92	1.00	0.83	0.58	0.83	0.83
D	0.00	0.08	0.17	0.33	0.42	0.58	0.42	0.25	0.58	0.25	0.67	0.75	0.83	1.00	0.75	0.92	0.92
E	0.25	0.33	0.42	0.08	0.17	0.25	0.17	0.50	0.33	0.50	0.42	0.50	0.58	0.75	1.00	0.75	0.75
F	0.00	0.08	0.17	0.33	0.42	0.50	0.42	0.33	0.58	0.33	0.67	0.75	0.83	0.92	0.75	1.00	0.92
G	0.00	0.08	0.17	0.33	0.42	0.50	0.42	0.33	0.58	0.33	0.67	0.75	0.83	0.92	0.75	0.92	1.00

Figure 3. (**a**) Dissimilarity and (**b**) similarity metric for LCZ classes (Appendix A in [4]).

Figure: LCZ (Local Climate Zone) based wights matrix reference

Spatial Hedging - Weight Matrix - Global index Y_g

- Model with no spatial dependency in Λ, α, and σ.
- Spatial dependency modeled by a spherical correlation function (Benth 2012)

$$q(x, y) = 1 - \frac{3}{2}\frac{|x-y|}{\gamma} + \frac{1}{2}\frac{|x-y|^3}{\gamma^3}$$

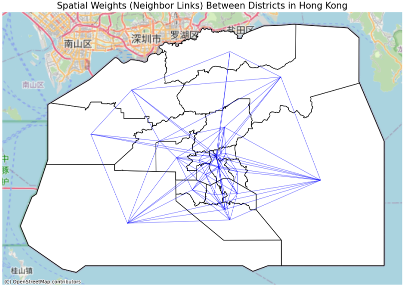

Figure: Rent Roll based